

birdclef-2026

BirdCLEF 2026 7th solution

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I know many competitors have scores above 0.955, so my solution probably doesn't have anything special. A lot of it might just be luck. Here I'll just share some thoughts and analysis from the competition, with some modelling details.

Data Preprocessing

Most of my time went into data work which is about quantity and quality.

The "Quantity" Problem

In this year's official data, there are three species with very few samples because their scientific names don't match between sources. They are:

- **Spot-tailed Nightjar** (*Anturus maculicaudus* ↔ *Hydropsalis maculicaudus*)
- **Ocellated Crane** (*Rufirallus schomburgkii* ↔ *Micropygia schomburgkii*)
- **Yellow-headed Caracara** (*Daptrius chimachima* ↔ *Milvago chimachima*)

The first name in each pair is what the official taxonomy and iNaturalist use, while the second is what xeno-canto uses. There are actually quite a few matching samples on xeno-canto for the official data.

The "Quality" Problem

A long-standing issue with BirdCLEF data is the mismatch in granularity between training data and the test set used for scoring. We know the test set is finely labeled by bird experts every 5 seconds. But the public data is mostly recorded by amateurs, with clips of all kinds of lengths (some over 5 minutes). The bigger problem is such long clips only get one label.

In this competition, around 4k+ samples have multi-class labels, meaning they have `secondary_labels`. Some samples even have more than 5 secondary labels. Imagine if you just randomly crop a 5/10s clip from the start or end. Can you really guarantee all those species are actually calling in that little window? I think it's very hard, and this introduces a lot of false positives. 4k+ samples is about 10% of the total, that's not a small number. So I chose to re-label these 4k+ samples myself.

My approach was to look at the spectrograms and try to pick out 10s clips with as many co-occurring species as possible. I use Audacity to view the spectrograms and 懂鸟app (aboutbirds) to help me recognize the species. If there are too many secondary labels (say >3) that I can't confirm, I'd cut out the false positives and only keep the ones I could verify. Sometimes one audio could be split into multiple samples. Let me use XC128801 as an example:

- Random-crop labels would be:
`grepot1` (primary_label) + [`'litnig1'`, `'whtdov'`, `'compau'`, `'chacha1'`] (secondary_labels)
- My choice:
17-27s clip → [`grepot1`, `'whtdov'`, `'compau'`, `'chacha1'`]
65-75s clip → [`grepot1`, `'litnig1'`, `'whtdov'`, `'compau'`, `'undtin1'`]

This way I got cleaner labels for rare classes like `chacha1` and `litnig1`. Along the way I also spotted a few labeling mistakes: XC2747 labels a weeping frog as `roahaw`; XC353371 mistakes `roahaw` for a Crested Oropendola; XC375114 is clearly `whbwar2`, not `flawar1`, etc. I also noticed that many recordings missed

labeling `redjun` (Red Junglefowl) — maybe these recorders just didn't think chickens were worth recording. Same thing happened with `47144` (Domestic Dog): some clips have dog barks but no label for them. These probably don't affect modelling much, just fun to share.

All this data work looks tedious and may not help modelling that much, but learning about different bird calls was honestly pretty fun.

Modeling

The modelling side isn't very different from last year, still `timm` backbone + SED head + 10s clips. The new thing this year is the Perch distillation approach, which helped me a lot. All my models use distillation, but I don't detach the backbone. The backbone is part of the gradient flow. I used two teachers for distillation: **Perch v2** and **AudioProtoPNet-5-BirdSet-XCL**. BirdSet is pretrained on 9k+ species, and in my experiments it gave better results with certain backbones. I also trained my own version of the open-source ProtoSSM + MLP model. The final solution is a rank ensemble of 4 SED models + 1 open-source-style model.

Model and Hyperparameter Choices

My final backbones are: `tf_efficientnetv2_s` with `Perch v2`, `mambaout_tiny` with `BirdSet`, `eca_nfnet_l0` with `BirdSet`. They score around 0.946-0.949 individually. `mambaout_tiny` has a really nice training loss curve, the only downside is slow inference. I tried the smaller `mambaout_kobe`, which was faster, but the score dropped about 0.003. Since I was only planning to use these three models, 90 minutes is more than enough for me, so I went with the slightly bigger model.

The other thing was mel parameter choice. Before I tuned the mel parameters, I can hardly break 0.92 (without pseudo labels). After settling on the parameters below, a single model could hit 0.94 (without pseudo labels):

- `n_fft = 1024` or `2048`, `n_mels = 128`, `hop_length = 320`, `fmin = 50`, `fmax = 16000`
- The final input to the model is shape `(128, 512)` (`320000 / 320 = 1000` downsample to 512).

Epochs are around 30, learning rate `1e-3`. I keep distillation off for the first few epochs and turn on Perch or BirdSet distillation from the 6th or 8th epoch. Since I use 10s as input, so I have to randomly cut 5s from each 10s to feed into the distillation teacher in every batch. I use model soup for inference on kaggle, such as combining the checkpoints from `[24, 28, 32]` epoch.

Using Pseudo Labels

After generating pseudo labels, I just concat them into the main hard-sample data. But this would mess up the label distribution and make the training set too huge. My fix is, in each epoch, I randomly weight-sample the pseudo-label portion (kind of like last year's `top-1 ss_weight`). This keeps things random so the model sees a diverse set of samples, while also keeping training size manageable to save time. In the end about 10k pseudo samples get mixed in with the hard-sample main data each epoch. I only did one round of pseudo labelling.

Training Details

Through EDA I noticed that most soundscape clips contain **nocturnal** species. To help the model learn these nighttime co-occurrence patterns better, I set my mixup so that nocturnal samples get mixed with other

nocturnal samples with high probability (0.8).
For very rare classes I upsampled them to about 20-30.

The Open-Source Model

Honestly this model did help me, but the open-source version has serious overfitting risk. I only kept its ProtoSSM + MLP structure, and threw out the site-hour-prior, residual model, and that huge pile of post-processing. The training data for ProtoSSM + MLP was the 66 labeled soundscapes + my own pseudo labels. One more thing about the Perch species mapping: in our data, 1595929 corresponds to the scientific name **Lysapsus limellum**, but Perch's matching name is **Pseudis limellum**. If you handle this step, predictions for 1595929 should be better. Adding my own open-source-style model gave me about +0.005 improvement.

That's about all that's worth sharing. I am very happy for winning a gold medal.

Share you the Java Kingfisher I photographed in Bali, Indonesia. Hoping next year's BirdCLEF will introduce us to more amazing birds!